No detectable edge: cointegration pairs trading on the Russell 3000, 2023–2025

A pairs strategy with formation-only selection, next-day fills and a per-stock cost model, tested in a main backtest, in a continuous walk-forward and against two baselines. Its returns cannot be told apart from zero, from the textbook distance method or from random pairs. A pre-registered test on 2026 data is pending.

Navnoor Bawa · Note of 20 October 2026 · Study run v32.0.0 of 18 October 2026, commit 711b18c

Code and data github.com/NavnoorBawa/russell3000-pairs-trading
Research-note site navnoorbawa.github.io/russell3000-pairs-trading

Summary

NO DETECTABLE EDGE

We test whether a cointegration pairs-trading strategy on the Russell 3000 earns a return after realistic execution costs. From July 2023 to December 2025 the strategy returned −0.30% on its capital (Sharpe −0.07; 95% interval −1.39 to +1.23), and its continuous walk-forward out-of-sample Sharpe is −0.33 (−1.49 to +0.86). Neither differs from zero, from the Gatev distance method (difference −0.14, p = 0.879) or from random pairs (28th percentile). Before costs the average trade earned +11.4 bp of its gross notional (t = 0.37) against 12.5 bp of modelled cost, so there is no gross edge for costs to remove. The sample rules out a Sharpe ratio above 0.97 (one-sided 95% bound) but cannot tell 0 from 0.5. The decisive test, on untouched 2026 data, stays sealed until its pre-registration is filed.

HOLDOUT 2026 · NET SHARPE	WALK-FORWARD OOS SHARPE	DEFLATED SHARPE, OOS
Sealed	**−0.33**	**0.4%**
Until the pre-registration is filed and the v33 code frozen. No 2026 price has been fetched.	95% CI −1.49 to +0.86 · HAC p = 0.559 · 711 days	Probability the Sharpe beats the best of 32 null trials · P(SR > 0) = 28.7%
GROSS EDGE PER TRADE	**BREAK-EVEN COST MULTIPLE**	**DATA SNOOPING · SPA P**
+11.4 bp	**0.69×**	**Pending**
t = 0.37 · n = 161 · cost 12.5 bp per trade	Net P&L is zero at this fraction of modelled costs; the gross itself is not significant	Declared family: 27-point grid, 11 toggles; runs on the v33 development calendar

1 Introduction

1.1 The question

Pairs trading buys one stock and sells another when the spread between them is unusually wide, and closes the position when it narrows. Gatev et al. (2006) documented excess returns from a simple distance rule on US stocks over four decades; later studies find that the edge has shrunk and is small after realistic costs (Do and Faff 2010; 2012; Rad et al. 2016), and Krauss (2017) surveys the methods. This note asks one question and is built to answer it honestly: **does a cointegration pairs-trading strategy on the Russell 3000 have an edge after realistic execution and costs, out of sample?**

The strategy selects pairs with the Engle–Granger test (Engle and Granger 1987) on formation data only, tracks each pair's hedge ratio with a two-state Kalman filter (Kalman 1960), trades a z-score of the hedged spread with fills at the next day's close, and pays costs computed from each stock's own liquidity. It is evaluated in a main backtest from 3 July 2023 to 31 December 2025, in a continuous

walk-forward with 11 out-of-sample windows, and against two baselines run on the same footing: the distance method of Gatev et al. (2006) and 200 portfolios of random pairs.

The answer for 2023–2025 is a null. A null is informative only to the extent that the test had power and the measurement was right, so most of this note is about what the sample can and cannot rule out, and about the checks that the code measures what it claims. The development period cannot test the design choices made on it; a pre-registered test on 2026 data, sealed at the date of this note, does that (Section 10).

1.2 What is new

The study uses free data only. Academic studies avoid survivorship with paid point-in-time databases; a backtest on free data usually starts from today's index members instead, and is exposed to the other familiar weaknesses of backtests: flat cost assumptions, rolling windows that drop positions at their edges, significance without power, and an untold number of designs tried. Each item below addresses one of these; the last adds a test the development data cannot provide.

- **A point-in-time universe from free SEC filings.** The complete holdings of the iShares Russell 3000 ETF (IWV) are public on Form N-PORT for the last month of each fiscal quarter; 26 quarterly snapshots give index membership as it was at each formation date, without a paid constituent history (Section 2). The v33 design trades this universe; the v32 run reported here does not.
- **Costs from each stock's own liquidity.** Every fill pays commission, half the typical quoted spread for the stock's dollar-volume bucket and square-root market impact on the stock's own volatility and dollar volume, at the next day's close (Section 7).
- **A continuous walk-forward.** The out-of-sample windows are calendar slices of one simulation with the ranker refit per window; positions carry across window boundaries instead of being discarded (Section 3).
- **Power analysis.** Each null is reported with the Sharpe ratios it excludes, the Sharpe ratio it could have detected and the track record a given Sharpe ratio would need (Section 5).
- **A declared variant family.** 27 threshold settings and 11 design toggles were declared on 20 October 2026, before any variant ran, for White's Reality Check, Hansen's SPA test, the Romano–Wolf stepdown, the model confidence set and the probability of backtest overfitting (Section 5.3).
- **A pre-registered 2026 holdout.** Hypotheses, test, multiplicity correction and decision rule are fixed before any 2026 price is fetched; each strategy then runs once and its result is published whatever it is (Section 10).

1.3 Related work

The published record points one way: the classic pairs edge was large before the 2000s, has shrunk since, and is small after realistic costs (Table 1). The closest analogue of this study is Rad et al. (2016), who compare distance, cointegration and copula methods on US stocks; this study differs in period, cost model and inference. Avellaneda and Lee (2010) trade residuals of single stocks against sector ETFs and factors, the second strategy family tested here (Section 9).

TABLE 1

Published results for pairs trading and residual reversion, as reported by their authors

Study	Sample	Method	Reported result
Gatev et al. (2006)	US, 1962–2002	Distance method, top-20 pairs	Annualised excess returns of up to 11% before costs, falling in later years. The baseline replicated here.
Do and Faff (2010)	US, to 2009	Distance method	Profitability declines over time; the strategy does best in prolonged turbulence.
Avellaneda and Lee (2010)	US, 1997–2007	PCA and sector-ETF residual reversion	Net Sharpe 1.44 (PCA) and 1.1 (ETF) over 1997–2007; PCA only 0.9 over 2003–2007. The second family tested here.
Do and Faff (2012)	US, 1963–2009	Distance method with trading costs	About 30 bp a month for well-matched pairs in refined industries after costs; largely unprofitable after 2002.
Clegg (2014)	US	Persistence of cointegration	Little evidence that pairs cointegrated in one year stay cointegrated the next (as summarised by secondary sources).
Rad et al. (2016)	US, 1962–2014	Distance, cointegration and copula methods	91, 85 and 43 bp a month before costs; 38, 33 and 5 bp after; trading opportunities fall after 2009. The closest published analogue of this study.
Krauss (2017)	Review	Survey of pairs-trading approaches	Distance, cointegration, time-series, stochastic-control and other approaches compared; open problems include costs and robustness.

2 Data

Every input is free: no account, API key or licence is needed to rebuild the study (Table 2). Prices are not redistributed with the code, under the vendor's terms; each run's fetch manifest lists the symbols and dates a re-run fetches.

TABLE 2
Data sources, all free

Source	What it provides	Used for
Yahoo Finance chart API	Daily open, high, low, close and volume, splits and dividends	Prices of the universe, the 11 sector SPDR ETFs and the VIX
SEC EDGAR: Form N-PORT filings of IWV	Quarterly holdings of the iShares Russell 3000 ETF	Point-in-time index membership (v33)
OpenFIGI mapping API	CUSIP to ticker	Joining holdings to price histories
SEC EDGAR: company submissions	SIC industry code per registrant	Fama–French 49 industries for same-industry pairs (v33)
Kenneth French data library	Fama–French five factors, momentum, short- and long-term reversal, one-month T-bill	Factor attribution (v33)
Nasdaq stock screener	Market capitalisation of US-listed common stocks	The v32 universe
Nasdaq Trader symbol directory	Primary listing venue	Venue-specific cost references

2.1 The v32 universe

The universe of the v32 study run is the 3,000 largest US common stocks by market capitalisation on 18 October 2026: the Russell 3000 rule applied to the Nasdaq screener, not FTSE Russell's constituent list. Daily prices from 2 January 2020 to 31 December 2025 were fetched for 97.1% of it, and 2,752 names have price history in the study data. Level screens, the $5 price floor and dollar volume, use point-in-time prices, closes with later splits undone, so that a split after the formation date cannot decide earlier membership. The 300 most liquid names by point-in-time dollar volume enter the pair search.

This universe has survivorship bias. Companies that failed, delisted or were acquired before 18 October 2026 are absent, and so are the pairs that broke when one of them failed. The bias flatters a mean-reversion strategy, which makes a null result on this universe more convincing, not less (Section 11).

2.2 Point-in-time membership from N-PORT filings

A registered fund reports its complete portfolio holdings on Form N-PORT, and the report for the third month of each fiscal quarter is made public on EDGAR. IWV tracks the Russell 3000, so its filings give index membership as it was, quarter by quarter, at no cost. `scripts/build_pit_universe.py` reads the equity-common holdings of 26 snapshots from 30 September 2019 to 31 December 2025, 2,581 to 2,895 holdings each, and maps each holding's CUSIP to a ticker through OpenFIGI (Exhibit 1). A snapshot is in force from the date its holdings describe until the next one. Two caveats: IWV is a fund, not the index, and holds fewer names than the index; and additions and deletions within a quarter appear only at the next quarter end. Snapshots dated 1 January 2026 or later are listed but not downloaded: their holdings carry a market value, and so a 2026 price, for every member.

2.3 The survivorship gap

The survivorship gap at a formation date is the index weight of members a backtest cannot trade: holdings without a usable ticker, plus members with a ticker whose price history the vendor no longer serves, mostly failures and takeovers. The v32 universe has no membership to measure the gap against. From v33 on, every formation records the snapshot in force, its members, how many have price history, and the index weight of those that do not (Exhibit 2).

3 Method

Selection sees formation data only; every trading decision on day t uses data up to day t and fills on day $t + 1$; the evidence is computed after the run (Exhibit 3). Tests change future prices and check that no past decision moves.

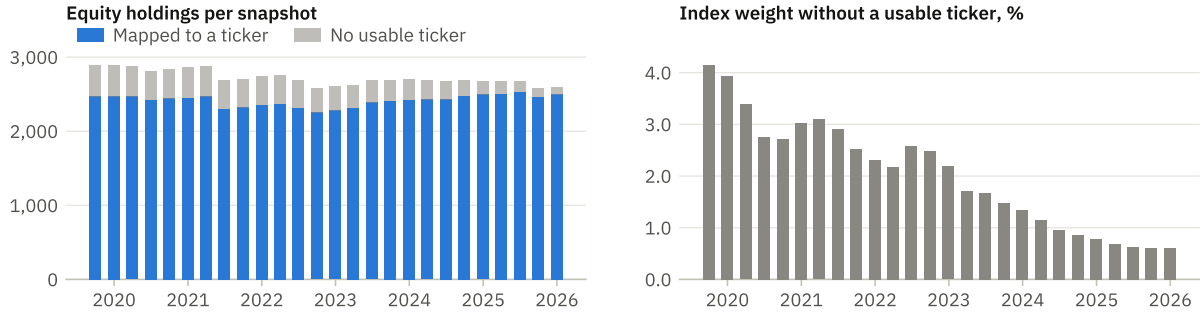
3.1 Formation

Pairs are formed once, on data to 31 December 2022. Among the 300 most liquid names, every pair with a positive return correlation of at least 0.25 is tested for cointegration with the two-step Engle–Granger

EXHIBIT 1

Free filings give 26 quarterly snapshots of index membership; the index weight without a usable ticker falls from 4.1% to 0.6%

Equity holdings of the iShares Russell 3000 ETF in its N-PORT filings, 2019–2025, and the share of their value that cannot be mapped to a ticker



Notes A holding without a usable ticker is one OpenFIGI cannot map, or one whose last ticker now belongs to another issuer and is left unmapped on purpose so that no member is joined to someone else's prices. The share is larger in older snapshots, as expected if most such holdings are securities that have since delisted; its weight is part of the survivorship gap.
Source `reference/universe/index.json`, built by `scripts/build_pit_universe.py` from SEC N-PORT-P filings (iShares Trust, series S000004341) and OpenFIGI.

EXHIBIT 2

Point-in-time universe: index members with and without price history at each formation date

Members of the index snapshot in force when pairs are formed, and the share of index weight the price data do not cover

Pending: v33 development run. The v32 universe is the 3,000 largest US stocks of 18 October 2026, back-filled to 2020, so it has no membership to report at a formation date. From v33 every formation records the index snapshot in force, its members, how many have price history, and the index weight of those that do not.

Notes The index weight of members without price history, plus that of holdings with no usable ticker, is the survivorship gap at that date: a pair that includes one of them cannot be formed.
Source The run's results export, `universe_by_formation`; membership from IWV's holdings in SEC N-PORT filings.

test in both directions, Bonferroni-adjusted across the two, at $p < 0.05$. A pair is kept if the half-life of its cointegrating residual lies between 4 and 25 trading days and a heuristic quality score passes: 1,776 of 35,491 pairs tested (Section 5.4). Every 90 trading days the pool is re-tested; pairs can drop out but none can enter.

3.2 Spread model

The hedge ratio is the slope state of a two-state Kalman filter on log prices,

$$\log p_{1,t} = \alpha_t + \beta_t \log p_{2,t} + \varepsilon_t, \quad (\alpha_t, \beta_t) \text{ a random walk},$$

with both log prices centred on their burn-in means and the filter seeded by ordinary least squares. The intercept makes β a hedge ratio rather than a ratio of price levels, and centring makes every signal

EXHIBIT 3

The pipeline, and what data each stage may see

Framed in the accent colour: the two look-ahead boundaries, the formation cutoff and the next-day fill

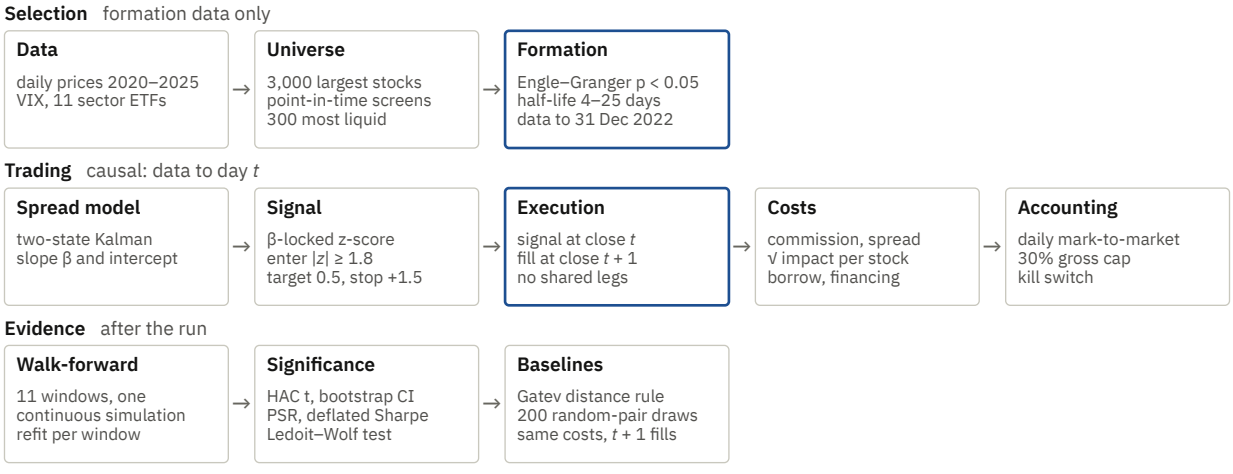
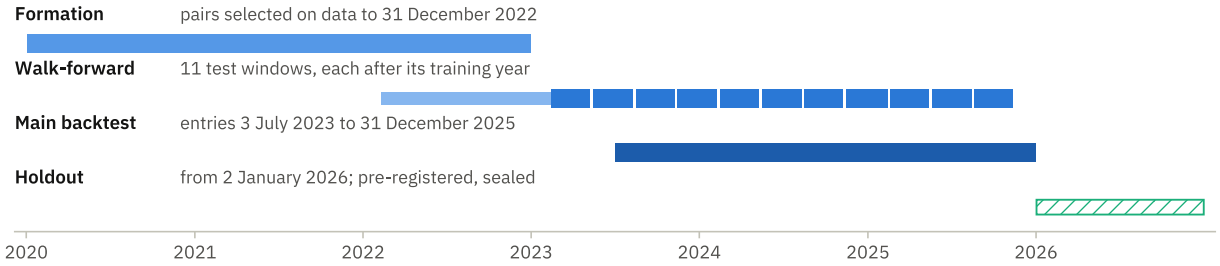


EXHIBIT 4

Data splits: formation, walk-forward, main backtest and the sealed holdout

Calendar 2020–2026



Notes The main backtest and the walk-forward overlap on 616 of the main backtest's 628 days and trade the same pair pool: they are one experiment, not two confirmations.
Source `results/v32_dev/pairs_trading_results_2023_2025_unbiased.json`; `pairs_trading/study_config.py` (calendar dev).

invariant to a vendor's back-adjustment of history for a later split or dividend, which shifts each log price by a constant.

3.3 Signal and exits

One z-score drives every decision. At a signal on day t_0 the hedge ratio is locked at its current value and the spread is standardised by its mean and standard deviation over the last 30 to 60 bars (three half-lives), both frozen at the signal:

$$s_t = \log p_{1,t} - \beta_{t_0} \log p_{2,t}, \quad z_t = \frac{s_t - \mu_{t_0}}{\sigma_{t_0}}.$$

A pair is entered when $|z| \geq 1.8$, long the spread when z is negative and short when it is positive. It exits at the target when $|z| \leq 0.5$, at the stop when z has moved 1.5 further against the position, and at the time stop after clamp(2.5 × half-life, 10, 25) trading days. Entries also require a 20-day return correlation of at least 0.20 and a hedge ratio in [0.25, 4]; a pair that has just lost money cools off before it can trade again; and a regime gate on the VIX and sector dispersion scales positions and blocks new entries in stressed markets.

3.4 Execution, accounting and risk limits

A signal computed on the close of day t is filled at the close of day $t + 1$, for entries and exits alike: in practice a market-on-close order in each stock's closing auction, so both legs print together. Share counts are fixed from the signal-day close and the legs are sized \$1 : \$β. Positions are marked to market daily, costs are booked on fill days, and positions still open at the end of the data are closed. The book has \$100M of notional capital, 3–10% of equity per pair, a 30% cap on gross exposure, at most 8 new entries a day, no stock in two open pairs, and drawdown and daily-loss kill switches that flatten it. Idle cash earns no interest.

A classifier of the probability that a trade is profitable, on 30 features, only orders same-day candidates, which matters when more signals fire than the book's limits allow; it does not create signals. Its out-of-sample AUC is 0.514 (0.510 pooled across the walk-forward windows; 0.5 is no skill), so it has no measurable skill, and v33 drops it from the primary design.

3.5 Walk-forward

The walk-forward is one continuous out-of-sample simulation on a 252-day training / 63-day test grid from February 2023 to December 2025. The ranker is refit on each window's training year, positions carry across window boundaries, and per-window figures are calendar slices of the one simulation (Table 17 in the appendix). It trades the same pair pool as the main backtest and shares 616 of its 628 days (Exhibit 4).

3.6 Inference

Sharpe ratios are annualised from daily net returns by $\sqrt{252}$. Means are tested with the HAC t-statistic of Newey and West (1987); intervals for single series are circular block bootstraps with the block length of Politis and White (2004) (5,000 resamples; blocks of 3 days for the main backtest and 2 for the

walk-forward). The probabilistic and deflated Sharpe ratios follow Bailey and López de Prado (2012) and Bailey and López de Prado (2014), using the variance of the Sharpe estimator under non-normal returns; differences of Sharpe ratios use the studentized bootstrap of Ledoit and Wolf (2008); and serial correlation enters the Sharpe ratio's standard error as in Lo (2002).

3.7 Validation

A backtest on real data says what a strategy did, not whether the code measures what it claims. `scripts/synthetic_validation.py` runs the real engine, with its full cost model, on synthetic markets whose answer is known (Table 3); continuous integration runs it on every push to the main branch and on pull requests. The test suite adds causality checks, each confirmed by re-introducing the bug it guards against: every decision dated on or before T is unchanged when all prices, volumes and macro data after T are replaced, with classical ranking and with a trained ranker; the regime gate's inputs are prefix-invariant; and the same trades occur at the same z-scores when the vendor back-adjusts history for a later split. The statistics are tested against reference implementations: statsmodels for Newey–West, arch for the Politis–White block length, an independent cross-validation for the probability of backtest overfitting, and the authors' code for Ledoit–Wolf.

TABLE 3
Known-answer checks

Check	What must happen	Result
Planted cointegrated pairs (OU spreads, half-life 5–14 days)	a large positive gross edge	*pending (committed validation run)*
Correlated, not cointegrated pairs (random-walk spread), forced into the universe	no edge; a positive mean would be look-ahead	*pending (committed validation run)*
Kalman hedge ratio against the true slope	recover it	*pending (committed validation run)*
Pair selection with planted pairs among unrelated names	find them; admit at most 5% of unrelated pairs	*pending (committed validation run)*

Notes Results fill from `results/validation/synthetic_validation.json` once a run is committed.

4 Development-period results

The main backtest enters trades from 3 July 2023 to 31 December 2025 on $100M of notional capital, with pairs selected on data to 31 December 2022. Its 161 trades returned −0.30% with a maximum drawdown of −3.00% (Exhibit 5). The book held no position on 207 of 628 days and used 15.1% of capital on average (gross), against a 30% cap.

The walk-forward tells the same story out of sample (Exhibit 6). 3 of 11 windows made money and the stitched path ends at −1.75%; the mean window return is −0.158% with a 95% interval from −0.636% to +0.320%. Two features of the main backtest deserve care. One month carries the result (Exhibit 7), and the half-year Sharpe ratios that look significant all contain that month (Exhibit 8); reading either as a regime the strategy could exploit would be fitting a rule to the test period.

5 Inference

Is the result significant? No: every interval in Exhibit 9 contains zero. The more useful question is what the data can rule out. With 628 daily observations the standard error of the annualised Sharpe ratio is 0.63, so the one-sided 95% upper bound is 0.97, and a true Sharpe ratio would need to be about 1.58 to be detected with 80% probability. A Sharpe ratio of 1.0 is rejected ($p = 0.046$); 0.5 is not ($p = 0.184$). A strategy whose true Sharpe ratio is 0.5 needs about 10.8 years of daily data to show as significant; at 1.0, about 2.7 years.

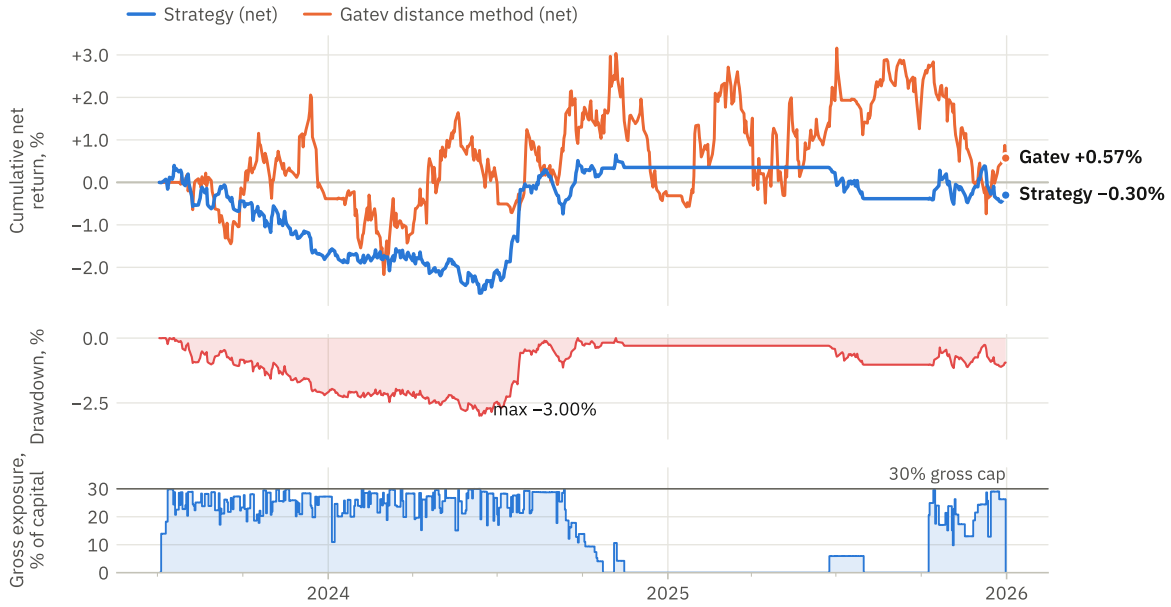
5.1 Power and the one-sided upper bound

A test that cannot reject zero is evidence of no edge only to the extent that it could have detected one. The power figures use the asymptotic variance of the Sharpe estimator for non-normal IID returns with the sample's skewness (0.17) and non-excess kurtosis (6.7) (Bailey and López de Prado 2012). At this sample length the critical Sharpe ratio of a one-sided 5% test is 1.04 and the Sharpe ratio detected with 80% probability is 1.58; the one-sided 95% upper bound, 0.97, is the largest Sharpe ratio the data do not reject (Table 4). The sample therefore excludes a strategy of Sharpe 1.5 or 2, is borderline at 1, and says nothing useful about 0.5.

EXHIBIT 5

Net of costs the book ends slightly below where it started; the Gatev baseline ends at +0.57%

Cumulative net return, underwater drawdown and gross exposure; main backtest, daily, July 2023 – December 2025



Notes No random-pair band: the v32 export holds the 200 draws' Sharpe ratios (Exhibit 10) but not their daily paths. Exports from v33 on carry each day's 5th–95th percentile of the draws' cumulative returns, and the band is then drawn here. The draws' mean path is not a substitute, because averaging 200 portfolios removes most of their noise. Gross exposure is rebuilt from the trades and reproduces the run's average and maximum (29.99%).
Source `results/v32_dev/pairs_trading_results_2023_2025_unbiased.json` (run of commit 711b18c); Gatev et al. (2006) distance method with the same $t + 1$ fills and cost model.

TABLE 4

What the development sample rules out

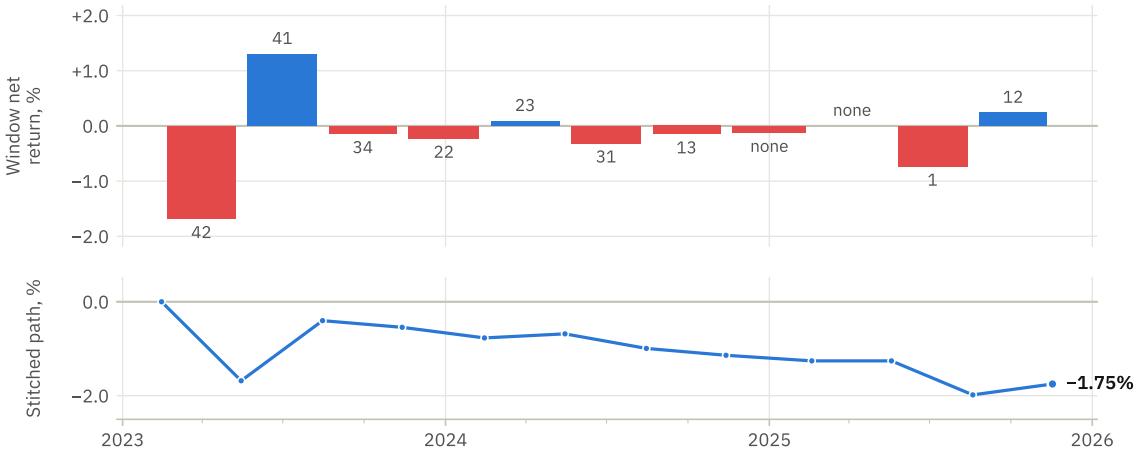
True annual Sharpe ratio x	p-value of H₀: SR ≥ x	Years of data for SR = x to be significant
0.5	0.184	10.8
1.0	0.046	2.7
1.5	0.007	1.2
2.0	0.001	0.7

Notes p-values are one-sided, evaluated at the observed Sharpe ratio. The track-record length is the minimum track record length of Bailey and López de Prado (2012) at the one-sided 5% level.
Source `assets/study.json`, `dev.power`.

EXHIBIT 6

Out of sample, 3 of 11 walk-forward windows made money; the stitched path ends at −1.75%

Net return per 63-day test window (bars, labelled with trade counts) and the compounded path at window ends; February 2023 – December 2025



Notes The windows are calendar slices of one continuous simulation with the ranker refit per window; positions carry across window boundaries. 2 windows made no trade. 11 windows have little power: mean −0.158% per window, 95% CI −0.636% to +0.320%, p = 0.478. The daily out-of-sample series is exported from v33 on; here the path is shown at window resolution.
Source `results/v32_dev`, `walk_forward_validation.windows`.

EXHIBIT 7

One month carries the result: July 2024 earned +2.14%, more than the whole period

Monthly net return, % of capital; the year column compounds the months shown

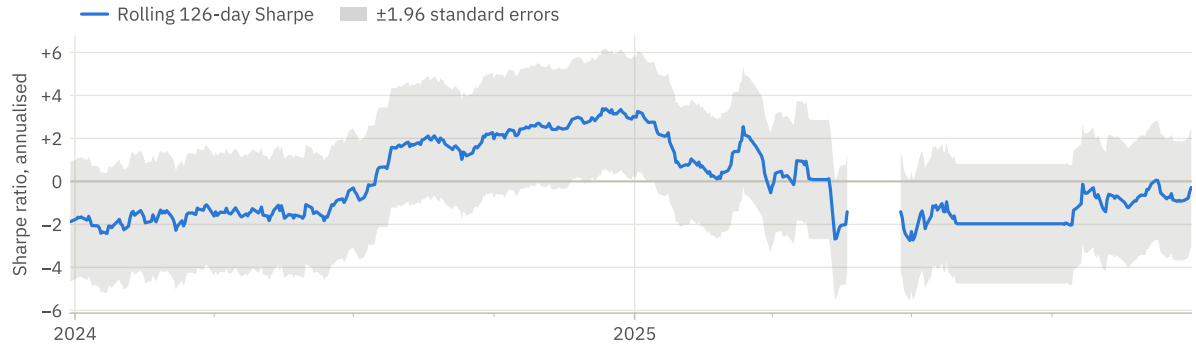
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
2023							−0.08	−0.35	−0.01	−0.74	−0.12	−0.48	−1.78
2024	+0.10	−0.15	+0.11	−0.46	+0.04	−0.01	**+2.14**	−0.01	+0.41	−0.04	+0.02	0.00	**+2.17**
2025	0.00	0.00	0.00	0.00	0.00	−0.12	−0.61	0.00	0.00	+0.41	−0.12	−0.21	−0.65

Notes 7 of 30 months were positive. The book was flat from December 2024 to May 2025; the regime gate blocked new entries on 248 of the 628 days. This grid is the evaluated period: a rule fitted to it would be data snooping.
Source `results/v32_dev`, `monthly_performance`.

EXHIBIT 8

Half-year Sharpe ratios swing between −2.8 and +3.4; every window whose band clears zero contains July 2024

Rolling 126-day Sharpe ratio, annualised, with a ±1.96 standard-error band; main backtest



Notes SE = √((1 + SR²/2)/*n*) per day with *n* = 126, annualised (Lo 2002), assuming IID returns. 30 of 503 overlapping windows clear zero, all ending between 21 November 2024 and 8 January 2025; overlapping windows are not independent tests. Windows in which the book was flat throughout have no Sharpe ratio.
Source `results/v32_dev` daily net returns; computed by `scripts/build_site_data.py`.

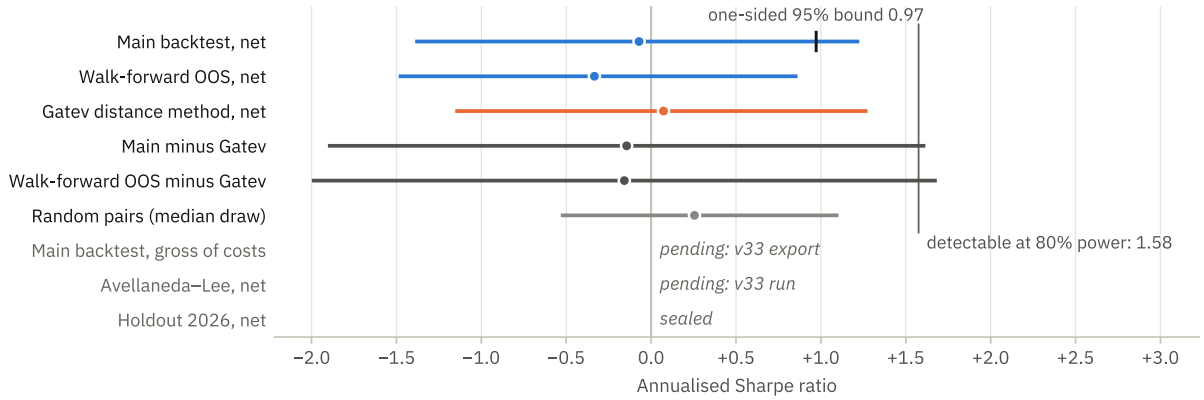
5.2 Baselines

The Gatev distance method, run on the same universe and period with the same $t + 1$ fills and cost model, returned +0.57% (Sharpe +0.07; 95% interval −1.15 to +1.27). The difference in Sharpe ratios

EXHIBIT 9

Every Sharpe interval spans zero; the data exclude only Sharpe ratios above about 1

Annualised Sharpe ratio with its 95% interval; development period, net of costs unless stated

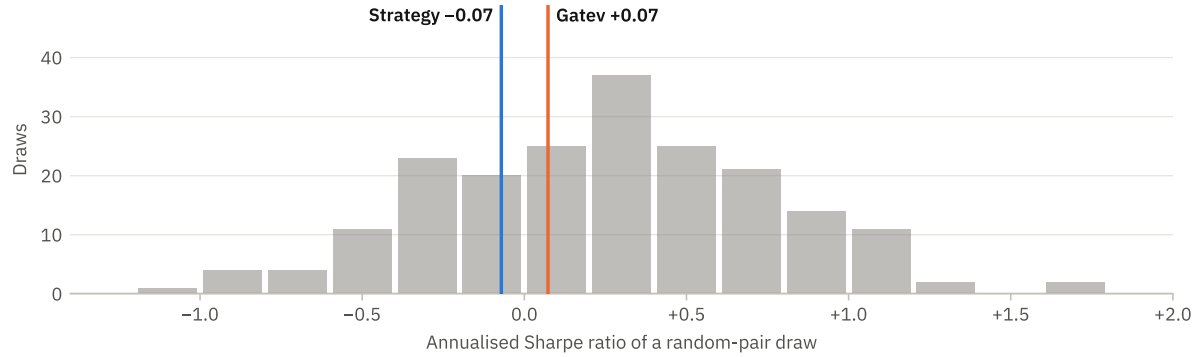


Notes Intervals are circular block bootstraps (Politis–White block length, 5,000 resamples) for single series and Ledoit–Wolf studentized bootstraps for differences; the random-pair row shows the 5th–95th percentile spread of 200 draws, not a confidence interval. The Gatev interval is computed by the site builder with the run's own bootstrap. Power and the upper bound assume IID returns with the sample's skew and kurtosis. Gross-of-cost, Avellaneda–Lee and holdout rows fill when their runs exist.
Source `results/v32_dev`, `significance` and `benchmarks`; `scripts/build_site_data.py`.

EXHIBIT 10

The strategy sits at the 28th percentile of 200 random-pair portfolios

Annualised Sharpe ratio of random 20-pair draws traded with the distance rule under the same costs; main period



Notes Median draw 0.26, 5th–95th percentile −0.53 to 1.10. The random control trades random pairs with the Gatev distance rule, not this strategy's engine, so it is a baseline for the distance method and only a rough one for this strategy; a control on the strategy's own engine is planned.
Source `results/v32_dev`, `benchmarks.random_control.draw_sharpes`.

is −0.14 in the main backtest (Ledoit–Wolf p = 0.879) and −0.16 out of sample (p = 0.859). Against 200 portfolios of random pairs the strategy sits at the 28th percentile (Exhibit 10).

TABLE 5

Significance tests, v32 study run of 18 October 2026

Test	Main backtest	Walk-forward OOS (daily)
Annualised Sharpe ratio	−0.07	−0.33
Newey–West t (mean ≠ 0)	−0.10	−0.58
Newey–West p	0.917	0.559
Probabilistic Sharpe, P(SR > 0)	45.5%	28.7%
Deflated Sharpe (32 trials)	1.4%	0.4%
Block-bootstrap 95% interval	[−1.39, +1.23]	[−1.49, +0.86]
Gatev distance method, Sharpe [95% interval]	+0.07 [−1.15, +1.27]	—
Strategy minus Gatev (Ledoit–Wolf p)	−0.14 (0.879)	—

Notes Newey–West t and p test a zero mean of daily net returns. The deflated Sharpe ratio is the probability that the Sharpe ratio exceeds the expected maximum of the stated number of null trials.

5.3 Deflated Sharpe and data snooping

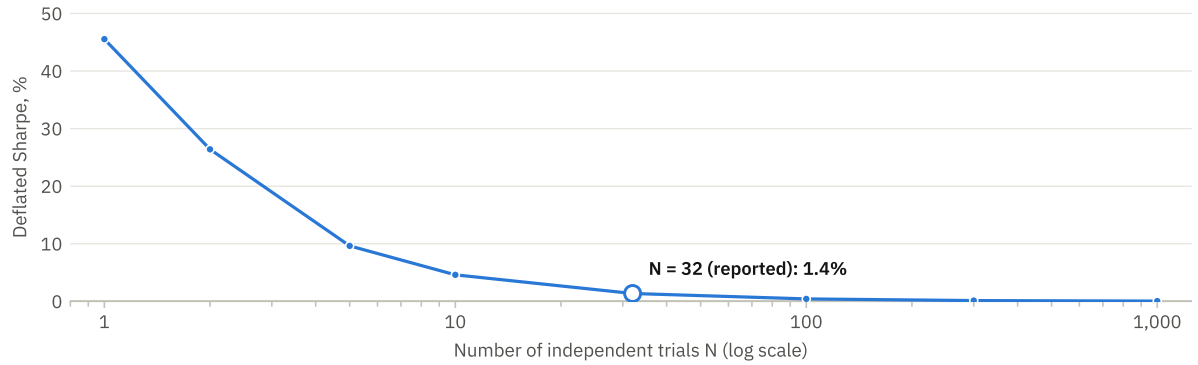
Every design choice in this study was made on 2020–2025 data over 32 versions. The deflated Sharpe ratio (Bailey and López de Prado 2014) asks whether the observed Sharpe ratio beats the best of N independent null trials; the run reports $N = 32$, a lower bound on the designs tried. At that N the main backtest has a 1.4% probability of beating chance and the walk-forward 0.4% (Exhibit 11). Harvey et al. (2016) make the same point for the factor literature, arguing that a new factor's t-statistic should clear about 3 rather than 2.

N is not known, and the deflated Sharpe ratio is only as good as the guess. The data-snooping tests replace the guess with a declared family. On 20 October 2026, before any variant ran, the study declared a 27-point grid of the entry, exit and stop thresholds and 11 design toggles around the v33 design (Table 16 in the appendix); each variant runs once on the development calendar and the primary design is not re-chosen from the results. The Reality Check of White (2000) and the SPA test of Hansen (2005) ask whether the best variant beats cash; the stepdown of Romano and Wolf (2005) identifies which variants do, with familywise error control; the model confidence set of Hansen et al. (2011) keeps the variants that cannot be told apart from the best; and the probability of backtest overfitting (Bailey et al. 2017) measures how often the in-sample winner falls below the median out of sample. Resampling uses the stationary bootstrap (Politis and Romano 1994).

EXHIBIT 11

Deflated for 32 trials, the main backtest's Sharpe ratio has a 1.4% probability of beating chance; the data-snooping tests are pending

Deflated Sharpe ratio of the main backtest against the number of independent trials *N* (log scale)



Notes *N* = 1 is the probabilistic Sharpe ratio; the run reports *N* = 32, a lower bound on the designs tried over 32 versions. The trial Sharpe variance is the null variance of the estimator at this sample length.
Source `pairs_trading.significance.deflated_sharpe_ratio` on `results/v32_dev` daily returns.

TABLE 6

Data-snooping tests over the declared variant family

Test	Result
Hansen SPA p (consistent)	*pending (v33 development run)*
White Reality Check p	*pending (v33 development run)*
Romano–Wolf: variants rejected at 5%	*pending (v33 development run)*
Model confidence set (90%): variants kept	*pending (v33 development run)*
Effective number of trials	*pending (v33 development run)*
Probability of backtest overfitting	*pending (v33 development run)*

Notes The family was declared on 20 October 2026, before any variant was run (`results/variant_family/declaration.json`); the tests run on the v33 development calendar.

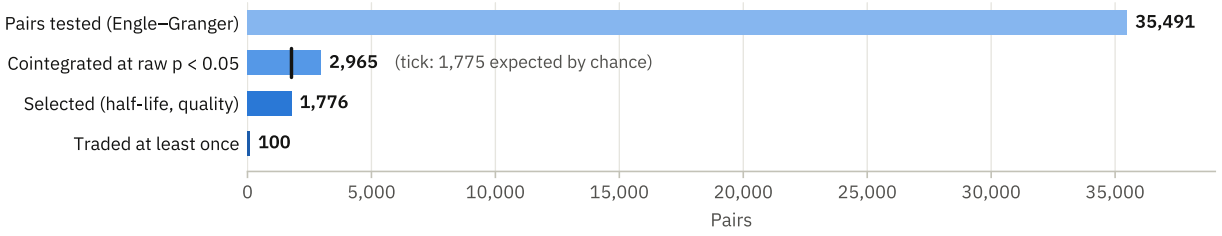
5.4 Multiple testing in pair selection

Pair selection is itself a multiple test. Of 35,491 pairs tested, 2,965 were "cointegrated" at $p < 0.05$, against 1,775 expected by chance alone. Only 1 survives a Benjamini–Hochberg correction at $q = 0.05$ (Benjamini and Hochberg 1995): the excess is spread thinly over many weak relationships. Selection kept 1,776 pairs, of which 100 were traded (Exhibit 12).

EXHIBIT 12

From 35,491 candidate pairs to 100 traded; chance alone accounts for most "cointegrated" pairs

Pairs at each selection stage, formation data to 31 December 2022



Notes Pairs share legs, so the tests are dependent; Benjamini–Hochberg is a diagnostic here, not a filter. Selection also requires a half-life of 4–25 days and a quality score.
Source `results/v32_dev`, `fdr_diagnostic` and `all_trades`.

EXHIBIT 13

Engle–Granger p-values against the uniform null

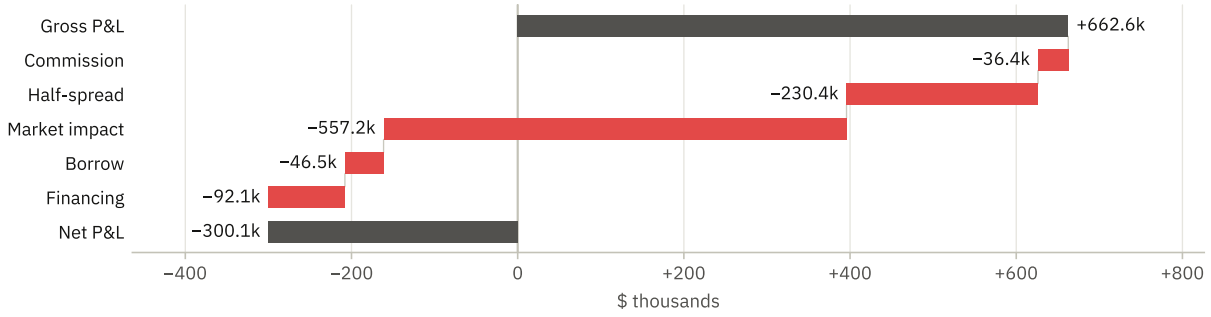
Histogram, 20 bins, with the height expected if no pair were cointegrated

Pending: v33 development run. The v32 export keeps the counts above but not the p-values themselves.

EXHIBIT 14

Gross P&L, itself not significant, does not cover modelled costs; market impact is the largest item

Implementation shortfall of the main backtest: gross P&L to net P&L by cost component, $ thousands



Notes The bars close exactly on the change in the book's value. Gross is 66.3 bp of starting capital, costs 96.3 bp, net −30.0 bp; per round trip, costs are 12.93 bp of traded gross notional. The signal-close paper P&L and the cost of the one-day delay are separated from v33 on.
Source `results/v32_dev`, `all_trades[].cost_breakdown_dollars` and the final portfolio value.

6 Where the edge goes

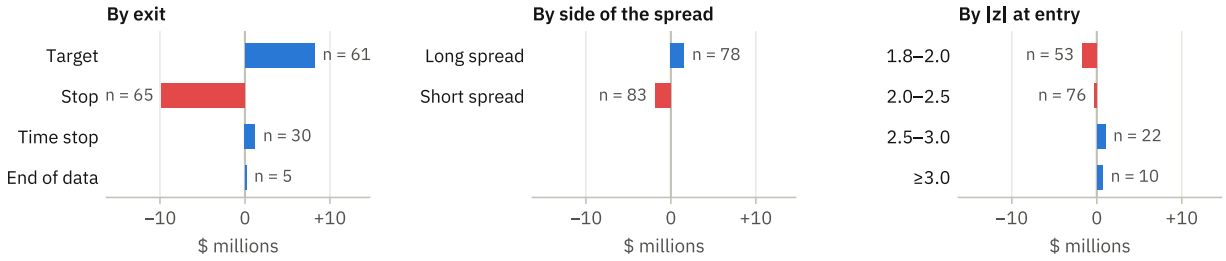
There is no gross edge for costs to remove. Before costs the 161 trades earned $662.6k; modelled costs were $962.7k, 57.9% of it market impact (Exhibit 14). Costs would have to fall to 0.69× their modelled level for net P&L to break even, and the gross return per trade is itself indistinguishable from zero (t = 0.37). The trades show the mechanism: the 65 stop-loss exits lost $9.80M, more than the 61 target exits earned ($8.18M; Exhibit 15).

The anatomy the evidence layer adds from v33 on is pending (Exhibits 16–19): the Sharpe ratio against a multiple of modelled costs, which needs each day's gross and cost split; the race between target and stop exits against a spread with no mean reversion; the decay of the same rule on the same pairs from formation to trading; and whether selected pairs stay cointegrated, the question Clegg (2014) studies (Table 1).

EXHIBIT 15

Stops lost more than targets earned, and short-spread trades lost about what long-spread trades made

Net P&L of the 161 trades by exit reason, side of the spread and |z| at entry, $ millions



Notes Long-spread trades made $1.55M and short-spread trades lost $1.85M, consistent with the net market exposure that hedge ratios below one leave in each trade (Section 8). Win rate 47.2% (76 of 161; 95% interval 39.6–54.9%). The |z| buckets describe the past and are not a selection rule: a higher entry threshold read off this panel would be fitted to the test period. Buckets use the exported z-score (three decimals).
Source `results/v32_dev`, `all_trades`.

EXHIBIT 16

Sharpe ratio against a multiple of modelled costs

0× to 2× costs, with the break-even multiple marked

Pending: v33 development run. Needs each day's gross and cost split, which the v32 export does not separate. On trade sums the break-even multiple is 0.69×.

EXHIBIT 17

Exit race: targets against stops, and against a spread with no mean reversion

Observed exits and the count expected under the null

Pending: evidence layer. Fills from the run's `anatomy.json` (`exit_race`).

EXHIBIT 18

In-sample to out-of-sample decay of the same rule on the same pairs

Gross bp per trade, formation period against the trading period

Pending: evidence layer. Fills from the run's `anatomy.json` (`decay`).

EXHIBIT 19

Do selected pairs stay cointegrated?

Share still cointegrated in later windows, against the 5% chance rate

Pending: evidence layer. Fills from the run's `anatomy.json` (`persistence`).

7 Execution and costs

Every fill pays its own costs, from the stock's own liquidity on that day, and fills at the next day's close: a signal seen at the close of day t trades at the close of day $t + 1$. The average trade held 12.6 trading days, and its larger leg was 1.01% of that stock's dollar ADV. Annual turnover was 6.0× capital. Table 7 sets out the cost model against published references and what v33 changes; `docs/EXECUTION_COSTS.md` compares the model with published impact estimates at each participation rate, and `tests/test_cost_model_zoo.py` keeps it inside that band.

TABLE 7

The v32 cost model against published references, and what v33 changes

Component	v32 model	Reference	v33
Commission	$0.002 per share, $0.50 minimum per order	Agency algorithmic rates are commonly quoted at 0.1–0.5¢ per share	unchanged
Half-spread	Half the typical quoted spread by 20-day dollar ADV: 0.75 bp (≥ $500M) to 40 bp (< $5M)	Frazzini et al. (2018)	bucket fit to be checked against measured spreads
Market impact	$Y \cdot \sigma_{daily} \cdot \sqrt{(\text{shares} / \text{ADV})}$, $Y = 0.3$, capped at 50 bp, on entry and exit	v32 cites Almgren et al. (2005) for $Y \approx 0.3$, which is their permanent-impact estimate; temporary-impact estimates run higher	Y stressed at 0.3, 0.6 and 1.0
Borrow	General-collateral fee, 25 bp a year on the short leg	D'Avolio (2002) on general-collateral fees; the v33 fee is a median buy-side indicative fee (`docs/EXECUTION_COSTS.md` gives the source)	one 0.38% fee a year replaces borrow plus financing
Financing	Prime-broker spread, 50 bp a year on the short half of gross	Overlaps with the borrow fee (a double count)	folded into the single fee
Fill	Close of $t + 1$ for entry and exit; legs $1 : $\beta	—	$t + 1$ open as a declared variant

Source `pairs_trading/transaction_costs.py`, `pairs_trading/study_config.py`; `docs/EXECUTION_COSTS.md`.

How much the conclusion depends on the cost model is measured, not asserted. Re-costing and re-sizing the main backtest's own trades under five other cost and sizing schedules (Exhibit 20, Table 8) turns none of them into a significant gain. A stress table, the impact coefficient at 0.3, 0.6 and 1.0 crossed with borrow at 0.38% and at a hard-to-borrow fee, is pending the v33 development run, as is a capacity curve (Exhibit 21).

TABLE 8

Cost schedules A–E: parameters and results of the replay

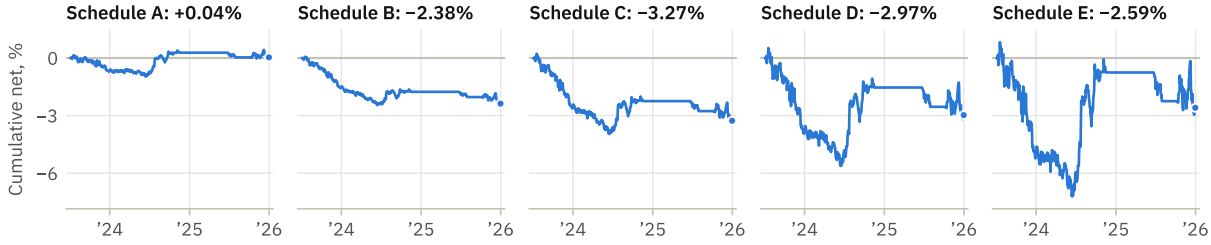
	Comm.	Spread	Impact cap	Borrow	Financing	Position	Return	Sharpe	Max DD	Costs	Peak gross	Trades
A	1.0	1.5	4	0.10	0.00	2–5	+0.04%	0.03	−1.09%	0.22	0.15	161
B	0.7	15.0	80	6.00	3.00	2–5	−2.38%	−1.58	−2.52%	2.57	0.15	152
C	5.0	10.0	50	2.00	1.50	4–10	−3.27%	−1.08	−4.17%	3.74	0.30	161
D	1.5	4.0	20	1.00	1.00	8–20	−2.97%	−0.48	−6.12%	3.88	0.60	161
E	1.0	2.0	12	0.60	0.50	12–30	−2.59%	−0.27	−7.96%	3.88	0.90	161

Notes Commission, half-spread and impact cap in bp; borrow and financing in % a year; position range in % of equity; return, maximum drawdown (Max DD) and total costs in % of capital; peak gross exposure as a multiple of capital; trades replayed.
Source `results/v32_dev/fund_type_comparison.json`; returns, Sharpe ratios and drawdowns as in `headline_metrics.json`.

EXHIBIT 20

Re-costed under five other cost schedules, the same trades lose money in four and break even in one

Cumulative net return, % of capital, when the main backtest's trades are re-sized and re-costed; schedules A–E in order of total cost

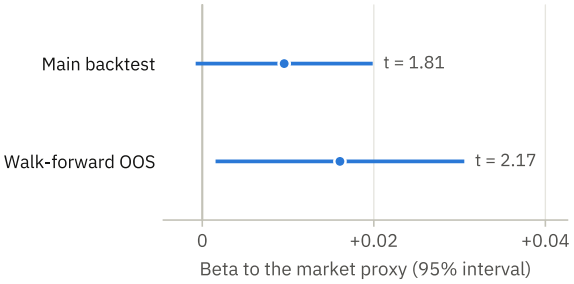


Notes Schedules are parameter sets, not descriptions of any firm; Table 8 lists each one's commission, half-spread, impact cap, borrow, financing and position size. Returns run from +0.04% (schedule A, the lowest costs and smallest positions) to −3.27% (schedule C). The replay keeps the main backtest's signals, entry and exit dates and daily paths, and changes sizes and costs; Table 8 lists the trades each schedule replays.
Source `results/v32_dev/fund_type_comparison.json`.

EXHIBIT 21
Capacity: impact cost against book size for a stated gross edge
Net bp per trade as notional scales

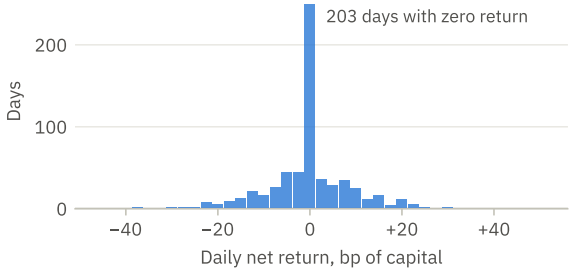
Pending: evidence layer. A break-even size computed on an insignificant gross edge would suggest an edge that has not been shown, so the curve is drawn for a stated hypothetical edge.

EXHIBIT 22
The only loading measured so far is a small market beta, significant out of sample
Beta to the sector-ETF market proxy with 95% interval (Newey–West)



Notes Proxy: equal-weight daily return of 11 sector SPDR ETFs. Factor rows for the five factors, momentum and reversal fill when the v33 export carries them.
Source `results/v32_dev`, `factor_attribution`.

EXHIBIT 23
Daily returns are fat-tailed (kurtosis 6.7) with a spike at zero from flat days
Distribution of daily net returns, bp of capital; main backtest



Notes Skew 0.17; kurtosis is non-excess (3 for a normal distribution). Fat tails widen the Sharpe ratio's standard error, which the power figures in Section 5 account for.
Source `results/v32_dev` daily net returns.

8 Risk and factor attribution

Market exposure is small but not zero. Against an equal-weight sector-ETF proxy the main backtest's beta is 0.010 (t = 1.81) and the walk-forward's 0.016 (t = 2.17) (Exhibit 22). The hedge ratio was below one in 148 of 161 trades, and with legs of \$1 and \$β such a trade is not dollar-neutral: 75 of 78 long-spread trades were net long and 74 of 83 short-spread trades net short. Annualised alpha is −0.25% in the main backtest (t = −0.24) and −0.84% out of sample (t = −0.80). The multi-factor attribution, on the five factors of Fama and French (2015), momentum (Carhart 1997) and short- and long-term reversal with Newey–West errors, arrives with v33.

9 A second strategy family: residual reversion

A null on one rule says little about the idea. Avellaneda and Lee (2010) trade each stock's residual against its sector ETF when that residual, modelled as a mean-reverting process, is stretched; many small concurrent bets give the test far more power than 161 pair trades. The parameters are taken from the paper unchanged: a 60-day regression on the sector ETF, an AR(1) fit of the cumulative residual, trades only when mean reversion is faster than 30 days, entry at $|s| > 1.25$, exits at −0.50 (long) and 0.75 (short), next-day fills and the same cost model. The arm runs on the development calendar with the v33 code and then, once, on the holdout (Section 10). It has not yet run on the development calendar, so Exhibit 24 is pending.

EXHIBIT 24
Avellaneda–Lee residual reversion against the pairs strategy, development period
Net Sharpe ratio with 95% interval, same universe and costs

Pending: v33 development run. Fills from `results/residual_dev/` when the arm has run on the development calendar.

10 The pre-registered 2026 test

Every design choice in this study was made on 2020–2025 data over 32 versions, so the development period cannot test those choices. The decisive test therefore runs on data none of them has seen: trading days from 2 January 2026. No 2026 stock price has been fetched. The protocol is committed first; each arm then runs once, and its result is published whatever it is. At the date of this note the pre-registration (`docs/PREREGISTRATION.md`) is not yet filed; Table 9 gives the hypotheses it fixes.

TABLE 9
Hypotheses of the pre-registration, filed before any 2026 price is fetched

	Arm	Hypothesis	Sharpe	p (Holm)	Decision
H1	v33 pairs (pre-registered design)	net mean return > 0	*sealed*	*sealed*	—
H2	Avellaneda–Lee residual reversion	net mean return > 0	*sealed*	*sealed*	—
H3	v32 as published, frozen	net mean return > 0	*sealed*	*sealed*	—

Notes Test: one-sided Newey–West t on daily net returns, Holm-adjusted across the three at 5%.

For each arm k the null hypothesis is that the mean daily net return over the holdout is at most zero, against the alternative that it is positive. The test is the one-sided Newey–West t-test on the daily net return series; the three p-values are adjusted with the step-down procedure of Holm (1979) at a familywise level of 5%, and an arm shows an edge out of sample only if its adjusted p-value is below 0.05. Whatever the outcome, each arm reports its net and gross return, its Sharpe ratio with standard error, the probabilistic Sharpe ratio, the block-bootstrap interval, the one-sided upper bound, trades and exposure, and, for the pairs arms, the Ledoit–Wolf difference against the Gatev method and the random-pair percentile on the same holdout (`scripts/evaluate_holdout.py`).

Power, stated in advance. About nine months of data can detect only a large Sharpe ratio. With returns shaped like the development period's (skewness 0.17, non-excess kurtosis 6.7, lag-1 autocorrelation 0.07), a one-sided 5% test with 80% power detects an annualised Sharpe ratio of about 3.1 in nine months and 2.7 in a full year (Table 10). The holdout is therefore a falsification check, not a proof of an edge: it can reject a large edge but cannot confirm a small one, and not rejecting the null is not evidence that the edge is zero, which is why each arm's upper confidence bound is reported. Running v32 alone first would spend the holdout, so v33 is frozen before any 2026 price is fetched, and all arms run once.

TABLE 10
Smallest annual Sharpe ratio the holdout detects with 80% power (one-sided 5% test)

Holdout length, trading days	Returns shaped like the development period's	IID normal returns
126	3.8	3.5
189	3.1	2.9
252	2.7	2.5

Notes About six, nine and twelve months. The development shape uses the asymptotic Sharpe variance for non-normal returns with lag-1 autocorrelation (`pairs_trading/power.py`); computed from development returns only.
Source `scripts/build_paper.py` on `results/v32_dev` daily net returns.

EXHIBIT 25
Cumulative net return on the 2026 holdout, by arm
Daily, from 2 January 2026; v33 pairs, Avellaneda–Lee and v32 frozen, same costs

Sealed. The pre-registration (`docs/PREREGISTRATION.md`) is committed before any 2026 price is fetched; this exhibit fills from `results/holdout2026/` after the single run.

11 Threats to validity

Each limitation, which way it biases the result, how large it is, and what addresses it (Table 11). Biases that flatter the strategy make the null more convincing, not less.

TABLE 11
Threats to validity of the v32 study run and the 2026 test

Threat	Direction of bias	Size	Mitigation
Survivorship, and an approximate index	↑ flatters	Unmeasured: the universe is today's 3,000 largest stocks, back-filled, so pairs broken by failures and delistings are missing	v33: point-in-time membership from IWV holdings in SEC N-PORT filings, with the coverage gap at each formation reported (Exhibit 2)
Design choices were made on the same data	↑ flatters	32 versions on 2020–2025; deflated Sharpe with 32 trials is a lower bound	Pre-registered 2026 holdout; declared variant family with SPA, stepdown, MCS and PBO
Re-selection can drop pairs but not add them	↓ penalises	The pool can only shrink after 2022	v33: full re-formation every 126 trading days on the trailing 3 years
Borrow is general-collateral for every name	↑ flatters	Borrow was $46.5k of $962.7k costs; specials and recalls are not modelled	v33: 0.38% buy-side median fee; hard-to-borrow stress
Fills are close-to-close	↕ either way	Larger leg 1.01% of ADV on average; no auction or queue model	$t+1$ open variant; closing-auction cost model planned

Threat	Direction of bias	Size	Mitigation
Ranker features include pair statistics estimated on the formation window	↑ flatters	Small: out-of-sample AUC 0.514, no skill to inflate	v33: the ranker becomes an ablation arm only
The "transformer" ranker sees one 30-feature vector	↕ none on P&L	A sequence of length one: functionally a residual MLP	v33: L2 logistic and gradient-boosted trees
Sector features are coarse	↓ penalises	Pairs are not restricted to one industry, so many are economically unrelated	v33: pairs within one Fama–French 49 industry
The 126-day re-selection check has little power	↓ penalises	It removes healthy pairs and keeps noise at similar rates (see the README)	v33: replaced by full re-formation
The price cache used through v31 was built with a full-sample penny-stock screen	↕ none for v32	The v32 run used a freshly fetched v32 cache	Fetch manifest and coverage check in every run
"Point-in-time" prices undo splits, not dividends	↕ small, either way	Early prices are understated by later dividends in the $5 price and dollar-volume screens	v33 keeps the raw close
Splits after the data window cannot be undone	↕ small, either way	The vendor back-adjusts history for every split up to the fetch date but reports split events only inside the requested window, so names that split after it are mis-screened; returns, z-scores and hedge ratios are unaffected (a constant factor)	Measured once the holdout opens and those split events can be fetched; disclosed
The regime gate was tuned on windows now inside the test period	↕ either way	New entries blocked on 248 of 628 days	v33: off by default; counted as a declared variant
Hedge ratios below one leave market exposure	↕ adds beta	Beta t = 2.17 out of sample (Exhibit 22)	v33: multi-factor attribution
The main backtest and the walk-forward overlap	↑ flatters	616 of 628 days in common, same pair pool	The holdout is the independent test
The random-pair control trades a different rule	↕ weakens the comparison	Random pairs run through the Gatev rule, not this engine	A control on the strategy's own engine is planned
The v32 arm's universe in the holdout is membership as of 18 October 2026	↑ flatters	Look-ahead in membership for the frozen v32 arm (H3) during 2026	Disclosed in the pre-registration; the v33 and Avellaneda–Lee arms use point-in-time membership
Factor files from the Kenneth French library extend into 2026	↕ none on trading	No strategy reads factor data	Every run truncates them at its own calendar's last date; no development export holds a 2026 factor return
Renamed securities with a new CUSIP look like delistings	↕ overstates the gap	Counted in the measured survivorship gap, which is therefore conservative	The data review remaps through N-PORT tickers and issuer successions

12 Reproducibility

Every number in this note is formatted by `scripts/build_paper.py` from `assets/study.json`, the file the research-note site's charts and checked numbers come from, and from the headline JSON that file was built from; the site's numbers are checked against the same JSON by `scripts/check_published_numbers.py`. A later price fetch will not match the cached data bit for bit, because vendors revise history; the run manifest (Table 12) pins what was used, and the sha256 of every committed source the note reads is listed in Table 13.

```
# the study (prices are fetched from Yahoo's chart API and cached)
pip install -r requirements.txt
python scripts/build_universe.py
python -m pairs_trading.main            # writes outputs/

# publish a run: copy outputs/ to results/<run>/, then
python scripts/build_site_data.py --dev results/v32_dev
python scripts/check_published_numbers.py --json results/v32_dev/headline_metrics.json

# this paper (needs the optional typst package: pip install -e ".[paper]")
python scripts/build_paper.py           # data, figures and PDF
python scripts/build_paper.py --check   # exit 1 if any of them is stale
```

Disclosures

Hypothetical, backtested results on $100M of notional capital, not achieved by any account. Returns are net of the modelled costs described in Section 7 unless labelled gross, earn no interest on cash, and are shown beside baselines run under the same costs. Hypothetical results have inherent limitations,

TABLE 12
Run manifest

Item	Value
Commit	`711b18cbd1a6 (clean)`
Run	`2026-10-18 09:21 UTC`
Price cache	`sha256 70776501502e08533cf17d46bbf466b26bb9d68233aa467aa1665d4ea5525ea1`
Headline JSON	`results/v32_dev/headline_metrics.json, sha256 93471564009554007e8bff1f6a67a02f7f2db96b99d76ad8bd3e5a57b5183b0e`
Site data	`assets/study.json, sha256 1d30c35b7ed630394e8528ef7c3f989750c77d95ecb435c1f969b190881c30f1`
Universe	`Nasdaq screener, 18 October 2026; 97.1% fetched`
Seed	`PAIRS_SEED=42`
Stack	`python 3.12.3 · numpy 1.26.4 · pandas 2.2.3 · scipy 1.15.2 · statsmodels 0.14.4 · torch 2.13.0+cpu`

TABLE 13
Committed sources read by this note

File	sha256
`results/v32_dev/fund_type_comparison.json`	`67c0da54b13d57570064fa4619d9505d4199786fb226b27ee86f4f6211249372`
`results/v32_dev/headline_metrics.json`	`93471564009554007e8bff1f6a67a02f7f2db96b99d76ad8bd3e5a57b5183b0e`
`results/v32_dev/pairs_trading_results_2023_2025_unbiased.json`	`0d2e376fdbac5de777aa33187938d9e0d23ef76207f4cdc92de93ce260f9e876`
`results/v32_dev/universe_meta.json`	`af2bc3d41d0b2cee0381c6479baa144d1a743cde7f78dc266de3bddac20116b5`
`results/variant_family/declaration.json`	`028b1a29935565931fbe191ac578d8f2756d36b1ddc1d7ecb4833b8f7ca66add`
`reference/universe/index.json`	`c64cc39c6431a8da4aee409914dd3a993f750ea42f71548c22b67ec35d398323`

Notes The PDF is compiled with a fixed timestamp, the note's date, so a rebuild of unchanged inputs is byte-identical; `paper/build/manifest.json` records the sha256 of every compile input and of the PDF.

including those listed in Section 11. This is a research project: not investment advice, not an offer of any product, and no claim of deployable performance. The cost schedules are parameter sets, not descriptions of any firm. Code under the MIT licence; IBM Plex typefaces under the SIL Open Font License 1.1.

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Appendix

A.1 Specification of the v32 study run

TABLE 14

Specification of the v32 study run

Stage	Rule
Universe	The 3,000 largest US common stocks by market cap on 18 October 2026 (the Russell 3000 rule applied to the Nasdaq screener), 97.1% fetched; daily prices 2020–2025; screens on point-in-time (split-unadjusted) price and dollar volume; the 300 most liquid enter the pair search.
Selection	Positive return correlation ≥ 0.25; two-direction Engle–Granger $p < 0.05$ (Bonferroni across directions); half-life 4–25 days on the cointegrating spread; quality score; re-selection every 90 trading days (re-test only; pairs can drop out, not enter).
Spread	Two-state Kalman filter (slope and intercept) on log prices centred on their burn-in means, so β is a hedge ratio and signals are invariant to vendor back-adjustment.
Rule	One β-locked z-score against its mean and standard deviation over the last 30–60 bars, frozen at the signal. Entry $\|z\| \geq 1.8$; target $\|z\| \leq 0.5$; stop 1.5 further against; time stop clamp(2.5 × half-life, 10, 25) days. Gates: 20-day correlation ≥ 0.20, β in [0.25, 4], loss cool-off, VIX and sector-dispersion regime scaling.
Book	$100M notional; 3–10% of equity per pair; 30% gross cap; at most 8 entries a day; no shared legs; daily mark-to-market; drawdown and daily-loss kill switch.
Ranker	A classifier of P(trade profitable) on 30 features, used only to order same-day candidates; out-of-sample AUC 0.514 (0.510 pooled across walk-forward windows; 0.5 is no skill).
Walk-forward	One continuous simulation on a 252-day train / 63-day test grid; the ranker is refit per window; per-window figures are calendar slices.
Evidence	Probabilistic and deflated Sharpe, Newey–West t, Politis–White block bootstrap, Lo (2002) adjustment, Ledoit–Wolf Sharpe-difference test, PBO; Gatev distance method and random pairs under the same costs and $t + 1$ fills.

A.2 Design changes in v33

TABLE 15

The v32 design and the pre-registered v33 design, from `pairs_trading/study_config.py`

Choice	v32 (run reported here)	v33 (pre-registered)
Universe	today's 3,000 largest US stocks by market cap	point-in-time members (IWV N-PORT)
Pair search	300 most liquid names, all pairs	500 most liquid names, same Fama–French 49 industry
Re-selection	re-test the 2022 pool every 90 trading days	full re-formation every 126 trading days on the trailing 3 years
Regime gate	on	off (a declared variant)
Ranker	learned classifier orders same-day candidates	none (classical ranking; ablation arm)
Borrow and financing	25 bp borrow + 50 bp financing a year	one 0.38% fee a year

Notes None of the v33 choices was made by its effect on development-period results; the reasons are stated next to each profile setting in the code.

A.3 The declared variant family

TABLE 16

Variants declared on 20 October 2026 for the data-snooping tests

Variant	What it changes
`ENTRY_Z`	grid values 1.50, 1.80, 2.10
`EXIT_Z`	grid values 0.25, 0.50, 0.75
`STOP_Z`	grid values 1.00, 1.50, 2.00
`regime_gate_on`	the v10/v11 gate fitted on now-test windows
`cusum_gate_on`	entry gate tried in earlier versions
`hurst_gate_on`	entry gate tried in earlier versions
`no_rolling_corr_floor`	decoupling filter (20-day correlation ≥ 0.20) removed
`ref_bars_40`	z-score reference window, shorter
`ref_bars_90`	z-score reference window, longer
`ranker_gbm`	the learned ranker (gradient-boosted trees) switched on
`open_fills`	t+1 open instead of t+1 close fills
`all_pairs`	no same-industry restriction
`universe_300`	v32's 300 most liquid names
`reform_every_63`	quarterly instead of half-yearly re-formation

Notes The threshold grid covers the three hand-set trading thresholds, which were chosen across versions on this same data. Each toggle changes one setting of the v33 design.
Source `results/variant_family/declaration.json`.

A.4 Walk-forward windows

TABLE 17

Walk-forward windows: calendar slices of one continuous simulation

Window	Training from	Test from	Test to	Net return	Sharpe	Trades	Ranker AUC
9	2022-02-11	2023-02-14	2023-05-15	−1.682%	−2.50	42	0.511
10	2022-05-13	2023-05-16	2023-08-15	+1.302%	2.39	41	0.506
11	2022-08-15	2023-08-16	2023-11-13	−0.143%	−0.27	34	0.557
12	2022-11-11	2023-11-14	2024-02-14	−0.227%	−0.44	22	0.565
13	2023-02-14	2024-02-15	2024-05-15	+0.088%	0.20	23	0.518
14	2023-05-16	2024-05-16	2024-08-15	−0.312%	−0.46	31	0.517
15	2023-08-16	2024-08-16	2024-11-13	−0.148%	−0.33	13	0.503
16	2023-11-14	2024-11-14	2025-02-18	−0.120%	—	0	0.525
17	2024-02-15	2025-02-19	2025-05-19	0.000%	—	0	0.560
18	2024-05-16	2025-05-20	2025-08-19	−0.733%	−2.80	1	0.487
19	2024-08-16	2025-08-20	2025-11-17	+0.236%	0.50	12	0.497

Notes Each window's ranker is trained on the year before its test start. Sharpe ratios of windows without a trade are not defined.
Source `results/v32_dev`, `walk_forward_validation.windows`.

A.5 Version history

TABLE 18

Selected versions; `docs/PROGRESS.md` **has the full record, deliberately unflattering**

Version	Change
v9–v12	First look-ahead fixes (P&L, pair selection, Kalman β drift).
v16–v19	Kalman spread fixes, β-weighted P&L, PCA residual cointegration, dynamic hold.
v24–v26.1	Ranker trained and wired in (ranking only); controlled ablation; a label bug that had disabled training.
v28–v29	Significance layer and baselines; $t + 1$ entry fills.
v30	Leakage audit: survivorship confirmed, selection-clean window tagging.
v31–v31.1	Pre-launch audit: inverted stop-loss, same-bar exits, future re-selections in the walk-forward; results withdrawn.
v32	Full audit and rebuild of the engine, Kalman model, accounting, walk-forward, costs, ranker, benchmarks and outputs; synthetic known-answer validation; study re-run: no detectable edge.
v33 (in progress)	Pre-registered design: point-in-time universe, same-industry pairs, full re-formation, regime gate off, one borrow fee, a second strategy family, the evidence layer and the sealed 2026 holdout.